Physics Worksheet: Capacitor Charge & Discharge

Square Signal Excitation (LFG) – MCQ Version

Name: ______________________ *Date:* ______________ *Grade 12*

Technical Context: Circuit Parameters

An electrical circuit is set up containing a Low-Frequency Generator (LFG) delivering a square wave voltage u_G, a resistor of resistance R, and a capacitor of capacitance C in series.
The LFG operates with a period T. During the first half-period ($0 \leq t \leq T/2$), the generator delivers a constant voltage E (Charging Phase). During the second half-period ($T/2 \leq t \leq T$), the voltage drops to 0 (Discharging Phase). The time constant of the circuit is $\tau = RC$.
The positive direction of current i is chosen to be the direction of the charging current.

Multiple Choice Questions

Circle the single best answer for each question.

1. Which equation correctly represents the law of addition of voltages for this series circuit at any instant t?

A. $u_G = u_C - u_R$
B. $u_C = u_G + u_R$
C. $u_G = u_C + u_R$
D. $u_R = u_G + u_C$

2. During the charging phase ($0 \leq t \leq T/2$), the generator voltage u_G is:

A. Steadily increasing from 0 to E.
B. Constant and equal to E.
C. Constant and equal to 0.
D. Decreasing from E to 0.

3. How does the voltage across the resistor, u_R, behave during the charging phase?

A. It remains constant at E.
B. It increases from 0 to E.
C. It decreases from E towards 0.
D. It remains constant at 0.

4. The relationship between the instantaneous current i and the voltage across the capacitor u_C is given by:

A. $i = \frac{1}{C}\frac{du_C}{dt}$
B. $i = C \cdot u_C$
C. $i = C\frac{du_C}{dt}$
D. $i = R \cdot C\frac{du_C}{dt}$

5. During the discharging phase ($T/2 \leq t \leq T$), what is the relationship between u_R and u_C?

A. $u_R = u_C$

B. $u_R = -u_C$

C. $u_R = E - u_C$

D. $u_R = 0$

6. What does a negative value for the voltage u_R during the discharging phase indicate about the current i?

A. The current has stopped flowing entirely.

B. The capacitor is gaining charge.

C. The current is flowing in the opposite direction to the charging phase.

D. The resistor is acting as a generator.

7. What is the SI unit of the time constant τ?

A. Farads (F)

B. Ohms (Ω)

C. Volts (V)

D. Seconds (s)

8. At $t = \tau$ during the charging phase, what approximate percentage of the maximum voltage E has the capacitor reached?

A. 50%

B. 63%

C. 99%

D. 37%

9. For the capacitor to be considered fully charged during the first half-period, which condition must the period T satisfy?

A. $T \geq 5\tau$

B. $T \leq 5\tau$

C. $T \geq 10\tau$

D. $T = \tau$

10. Based on the condition for a complete charge and discharge cycle, what is the maximum frequency f_{max} the LFG can be set to?

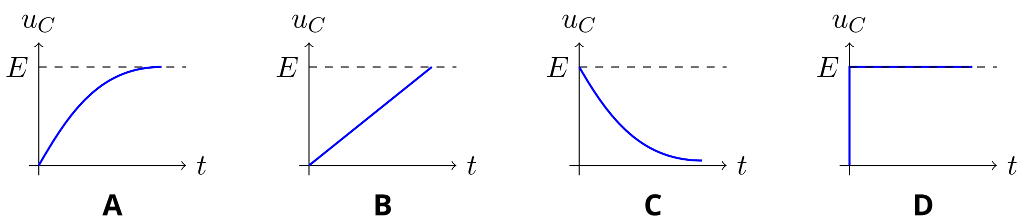
A. $f_{max} = \frac{1}{10RC}$

B. $f_{max} = \frac{1}{5RC}$

C. $f_{max} = 10RC$

D. $f_{max} = \frac{10}{RC}$

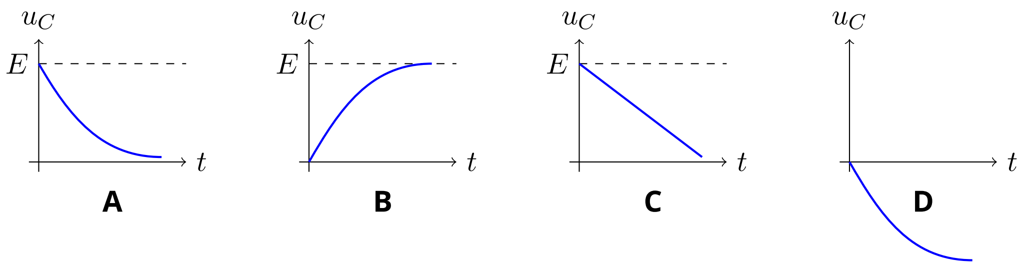
11. Which of the following graphs best represents the evolution of the capacitor voltage $u_C(t)$ during the charging phase?



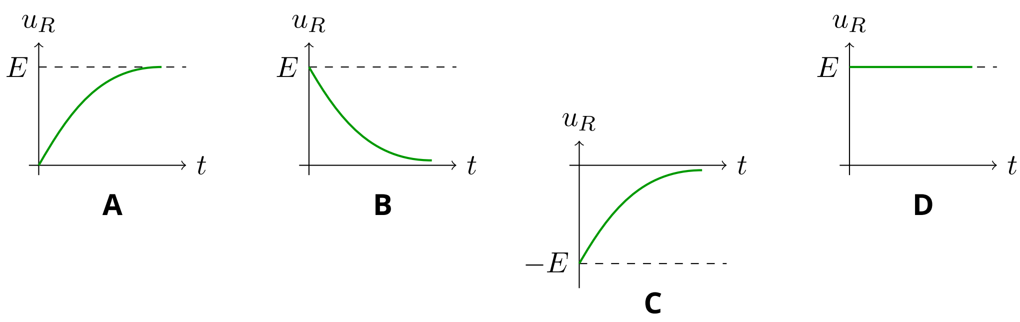
12. A student plots the experimental graph of $u_C(t)$ during the charging phase and draws the tangent line to the curve at $t = 0$. Which graphical method allows them to correctly determine the time constant τ?

A. Finding the abscissa where the voltage reaches exactly half of its maximum value ($E/2$).
B. Finding the abscissa of the intersection point between the tangent at $t = 0$ and the horizontal asymptote $u_C = E$.
C. Finding the abscissa where the tangent at $t = 0$ intersects the horizontal time axis.
D. Finding the time it takes for the capacitor to reach its maximum voltage E.

13. Which of the following graphs best represents the evolution of the capacitor voltage $u_C(t)$ during the **discharging** phase ($T/2 \leq t \leq T$)?



14. Which of the following graphs best represents the voltage across the resistor $u_R(t)$ during the **charging** phase ($0 \leq t \leq T/2$)?



15. Based on the relations between $u_R(t)$ and $i(t)$, which of the following accurately describes the evolution of the current $i(t)$ during the **discharging** phase?

A. It starts at a maximum positive value $\frac{E}{R}$ and decays exponentially to zero.
B. It is a constant negative value $-\frac{E}{R}$.
C. It starts at zero and decreases linearly to $-\frac{E}{R}$.
D. It starts at a maximum negative value $-\frac{E}{R}$ and approaches zero exponentially.

16. **(Bonus)** When using an oscilloscope with a common ground (Mass) to visualize u_C on Channel Y_1 and u_R on Channel Y_2 simultaneously (as shown in standard circuit diagrams), what must be done to accurately display u_R?

A. Add Channel 1 and Channel 2 together using the "MATH" function.
B. Press the "INVERT" (or reverse) button on Channel 2 because it measures $-u_R$.
C. Press the "INVERT" button on Channel 1 because it measures $-u_C$.
D. Change the time-base of the oscilloscope to clearly separate the signals.